

Mingtai Zhang

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Education

Guangxi University | School of Resources, Environment and Materials

2022 – 2026

B.Eng. in Materials Science and Engineering

Coursework in fundamentals of materials science, physical metallurgy, principles of metallurgical processes, materials characterization, physical chemistry, metal forming technology, mechanical design.

Research Summary

For my thesis I plate a thin magnesium layer onto lithium metal. Lithium spreads evenly on that layer. I assemble the cells myself, run the cycling myself, do the XRD work, do the EIS work. Every claim had to hold in three separate cell types before I believed it. My computational work asks the same two questions in another field. One is how many measurements are needed to fix an answer. The other is whether a reported number means what it claims. I wrote the data pipeline for a study on how few scans can locate defects. I wrote its software too. That imaging method also serves on electrode surfaces. I am first author on a large audit of machine-learning systems. I am applying for a PhD in computational materials, where the two sides are one job.

Research Interests

Lithium metal anodes, plating, stripping | separators coated with MXene, ion transport | imaging and machine learning for materials characterization | fast automatic screening of cells, labs that run their own experiments | metals recovered from waste, reused as catalysts

Publications, Manuscripts

Co-author lists for papers under double-blind review are withheld until notification, following the anonymity policy of each venue.

Materials, Electrochemistry

- [1] Guang Zhang, Dong Yang, Kewang Yang, Farwa Mushtaq, Meng Zhang, **Mingtai Zhang**, Jieshang Lu, Yucheng Wen, and Meinan Liu. “Modified Ether Electrolyte by Siloxane Additive with Abundant Si–O Bonds for High-Voltage Lithium Metal Batteries.” *Journal of Physics: Conference Series* **3165**, 012009 (2025); CEAMS 2025, Dalian [EI-indexed]. DOI 10.1088/1742-6596/3165/1/012009.
- [2] **Undergraduate thesis**. “Electrochemical Deposition of Magnesium for Constructing High-Performance Lithium Metal Anodes.” Guangxi University, 2026. Advised by Prof. Meinan Liu.

Measurement, Imaging, Machine Learning

- [3] Yuchen Miao, **Mingtai Zhang**, Zijun Wang, Ruobing Han, Chang Han, and Siyang Xu. “Two-Scan Recovery of Sparse Defects in Ptychography.” **Second author**; under review, *IEEE Signal Processing Letters*, 2026. (I did the data curation, wrote the software.)
- [4] Paper on attribution boundaries in conversational AI memory (title withheld during review) **First author**; under double-blind review, a top-tier HCI conference.
- [5] Paper on the accuracy–evaluation trade-off in test-time selection (title withheld during review) **Second author**; under double-blind review, a top-tier machine-learning conference.
- [6] Yuchen Miao, Zijun Wang, Chang Han, Yurui Shi, **Mingtai Zhang**, and Siyang Xu. “The ‘10th Juror’: Open-Set Standpoint Screening for Bureaucratic Bias Detection.” *EMNLP 2026* (Main Conference) [CORE A*; CCF-B].

Research Experience — Materials, Electrochemistry

Lithium-Metal Battery Group (Prof. Meinan Liu), Guangxi University | Undergraduate Researcher

2025 – present

Magnesium layers plated onto lithium foil | thesis project, sole author

- Built a coin cell that plates magnesium straight onto lithium foil. I held the current density fixed. I held the areal capacity fixed. Only the number of pre-cycles changed, 20, 30, or 40. All assembly was done in an argon glovebox with water, oxygen both below 0.1 ppm.
- The best result came at 30 pre-cycles. Each side of that point failed in its own way. At 20 pre-cycles the magnesium does not cover the whole surface. Bare lithium spots stay open, efficiency collapses at cycle 45. At 40 pre-cycles the layer grows too thick, cracks, pushes the overpotential to 180 mV. That cell collapses at cycle 30.
- The plating overpotential dropped from 145 to 114 mV at 1 mA per square centimetre. The nucleation dip dropped from about 30 to about 20 mV. The treated lithium held near 100% Coulombic efficiency against copper for 120 cycles. Untreated lithium fell from 98% to roughly 80%. Symmetric cells stayed flat near 30 mV for 330 h. Untreated lithium drifted from 80 to about 180 mV.
- Checked the same anode in a full cell against a commercial lithium iron phosphate cathode loaded at 9.9 mg per square centimetre. It kept 85.3% of its capacity after 600 cycles at 0.5 C. At 3 C it gave about 145 mAh per gram, 1.45 times the untreated cell. EIS showed the interface resistance falling from roughly 30 to 10 ohm. XRD showed magnesium metal plus a lithium–magnesium alloy phase. That phase gives the lower nucleation barrier. The route needs only commercial electrolyte, standard coin-cell parts. Other groups can repeat it.

Separators coated with MXene to slow down dendrites

- I run the whole MXene process myself. I etch the aluminium out of the parent ceramic powder, split the product into thin sheets suspended in liquid, coat those sheets onto the battery separator.
- The goal is a separator that does more than keep the two electrodes apart. Charged groups on the sheet surface hold the negative ions in place. More of the current is carried by lithium ions. Lithium reaches the anode more evenly. SEM, XPS, cycling tests link the surface chemistry to transport, to cycle life.

Siloxane additive for high-voltage ether electrolytes | JPCS 2025 (EI)

- Made the cells for this study, ran the electrochemical tests. The additive keeps ether electrolytes from breaking down above 4 V. Capacity retention rose from 70.8% to 73.9% over 700 cycles at 0.5 C. In the high-voltage cell it rose from 70% at 20 cycles to 75% at 100 cycles.

- Industrial wastewater carries many heavy metals at once. I turned that mixture into one single oxide phase. The waste stream becomes catalyst feedstock instead of sludge. I ran the synthesis, the phase control, the tests on breaking down volatile organic compounds under light. Oxides made from many metals at once already exist. Catalysts made from sludge already exist. Nobody has joined the two for this reaction yet. Real wastewater never gives a clean metal ratio. That makes the composition search a good target for machine learning.

Research Experience — Computation for Physical Measurement

The work above draws conclusions from measurements that never show the interface directly. The work below asks when such conclusions hold, what a proper check costs.

Locating Defects from Very Few Scans

2026

Second author, IEEE Signal Processing Letters, under review | simulation code, full test framework

- I did the data work, wrote the software. The study treats an imaging method that rebuilds a sample from its diffraction patterns. The sample is a flat known background carrying a few unknown defects. Defect checks on wafers, thin films, coatings pose the same problem.
- I built the simulator, the test framework. It covers beam shapes, scan paths, random defect sets, detector noise. I measured the new two-scan method against full-data linearization, against two standard iterative methods, over a range of defect spacings, a range of noise levels. Every operating point ran hundreds of trials.
- This project changed how I read my own lab data. I first ask whether the measurement can fix the answer at all. Which method reconstructs better comes second.

Large-Scale Computational Experiments

2025 – present

First author, under double-blind review | second author, under double-blind review | evaluation-set construction, EMNLP 2026 Main

- Led a large-scale audit of machine-learning systems. I wrote down one requirement that could be proved false. I split it into two separate measures. I tested it over hundreds of controlled episodes, tens of thousands of scored outputs, a large set of generated records in a multi-condition factorial design. A second model family served as a replication arm. One held-out text form marked the edge of the claim. The result was negative. I reported it that way. Two training methods improved one output channel. The same methods made the other channel worse.
- Second author on a paper about picking the best answer at test time. It measures how much accuracy stays reachable once one fixed pool of reused labels must also prove that accuracy. Better picking makes the outputs more alike, harder to tell apart. The paper states the trade-off, tests the resulting selection rules on four fixed reasoning benchmarks.
- Built the evaluation set for an EMNLP 2026 main-conference paper on agents that must cite their evidence. I also ran part of the analysis. The system reached an F1 score of 0.880, 20.2 points above the strongest baseline without training examples.
- Fast automatic screening of cells needs the same tools. Factorial designs. Thousands of scored trials. Replication arms. Stated limits on what a result covers. This is the part of my computational work I most want to bring back into materials research.

Industrial Experience

Four supervised placements, sixteen manufacturing enterprises across Guangxi

2024 – 2026

- Followed the aluminium chain from mine to product. Open-pit bauxite mining, ore blending, Bayer digestion, red-mud settling, calcination, molten-salt electrolysis to raw aluminium, casting, extrusion, rolling into sheet, rolling into foil. Sites included **Guangxi Huayin Aluminium** (Chalco JV, Debao), the **Chalco Guangxi Branch** at Pingguo, **Guangxi Huasheng New Material**, **Guangxi Guangtong Liuzhou Aluminium**.
- One site sits closest to my own research. At the former **Guangxi Nannan Aluminium Foil** works, now **Nanning Industrial Investment Aluminium-Based New Materials Group**, I followed the whole route from ingot to foil for lithium-ion battery aluminium. The thinnest product is 12 micrometre foil, bright on both faces. That foil carries the current in the cathode of a cell. The plant also ships carbon-coated foil, battery case stock, straight to electric vehicle battery lines.
- Also covered integrated steelmaking, stainless coil at **Guangxi Liuzhou Iron and Steel**, **Guangxi BG New Materials**; vehicle stamping, welding, painting, assembly at **Dongfeng Liuzhou Motor**; heavy structural fabrication at **LiuGong Machinery**; crankshaft die forging at **Guilin Fuda**; sintering of hard cutting materials under high pressure at **Guilin Tebang**; sand casting at **Guangxi Yuchai Foundry**; solar module lines at **Guangxi Dasheng New Energy**, **Guangxi Jingfei Energy**.

Ventures, Leadership, Honors, Technical Skills

Ventures. Founder, Kangjie tutoring service for students home on holiday, Yuncheng, Shanxi (winter 2026). University students taught for free. I ran the scheduling, matched tutors with students, tracked progress. AI agent tools did most of the coordination. The service reached 28 tutors, 37 high-school students. Co-founder, fitness coaching service with the Guangxi University Rehabilitation Association (2026 – present), now running live. I own the operating plan, the resources.

Leadership. Captain, Guangxi University Football Team | President, University Football Association | President, Student Union, School of Resources, Environment and Materials.

Honors. Silver Award, Guangxi University Career Planning Competition (2023) | Third Prize, Guangxi University Metallography Contest (2023) | First Place (2025), Third Place (2024), Guangxi “Qianli Cup” Intercollegiate Football League, undergraduate division.

Electrochemistry. Coin-cell assembly, argon glovebox, CV, EIS, electrochemical workstations | **Synthesis.** Electrochemical deposition, MXene etching, delamination, coating, oxides made from many metals at once, making electrodes, making cells | **Characterization.** XRD, SEM, XPS | **Computation.** Python, PyTorch, L^AT_EX, Origin; factorial design, test frameworks for very large experiments, physics-based simulation, statistical analysis | **Process engineering.** Bayer refining, electrolysis, casting, extrusion, rolling, forging, foundry, SOP, laboratory safety